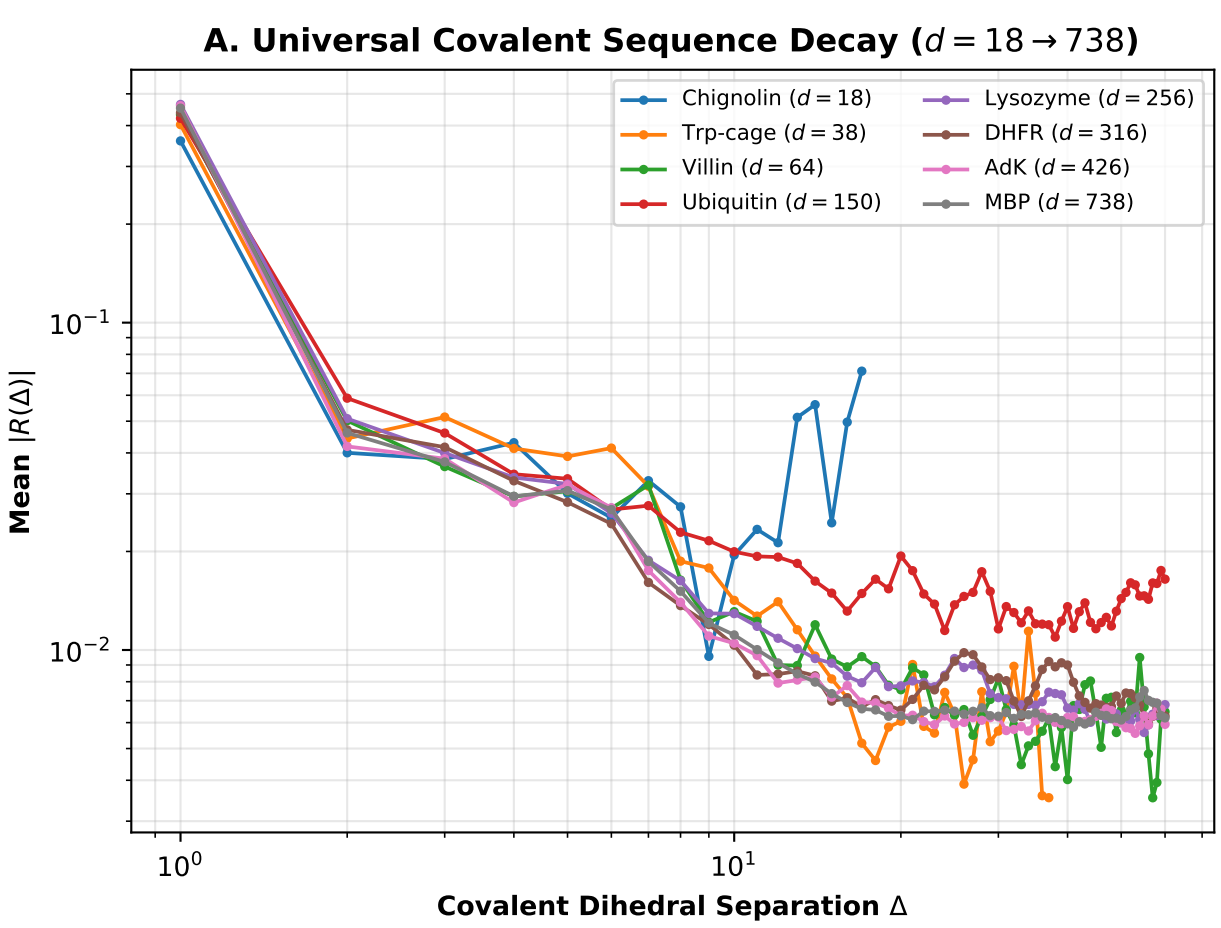
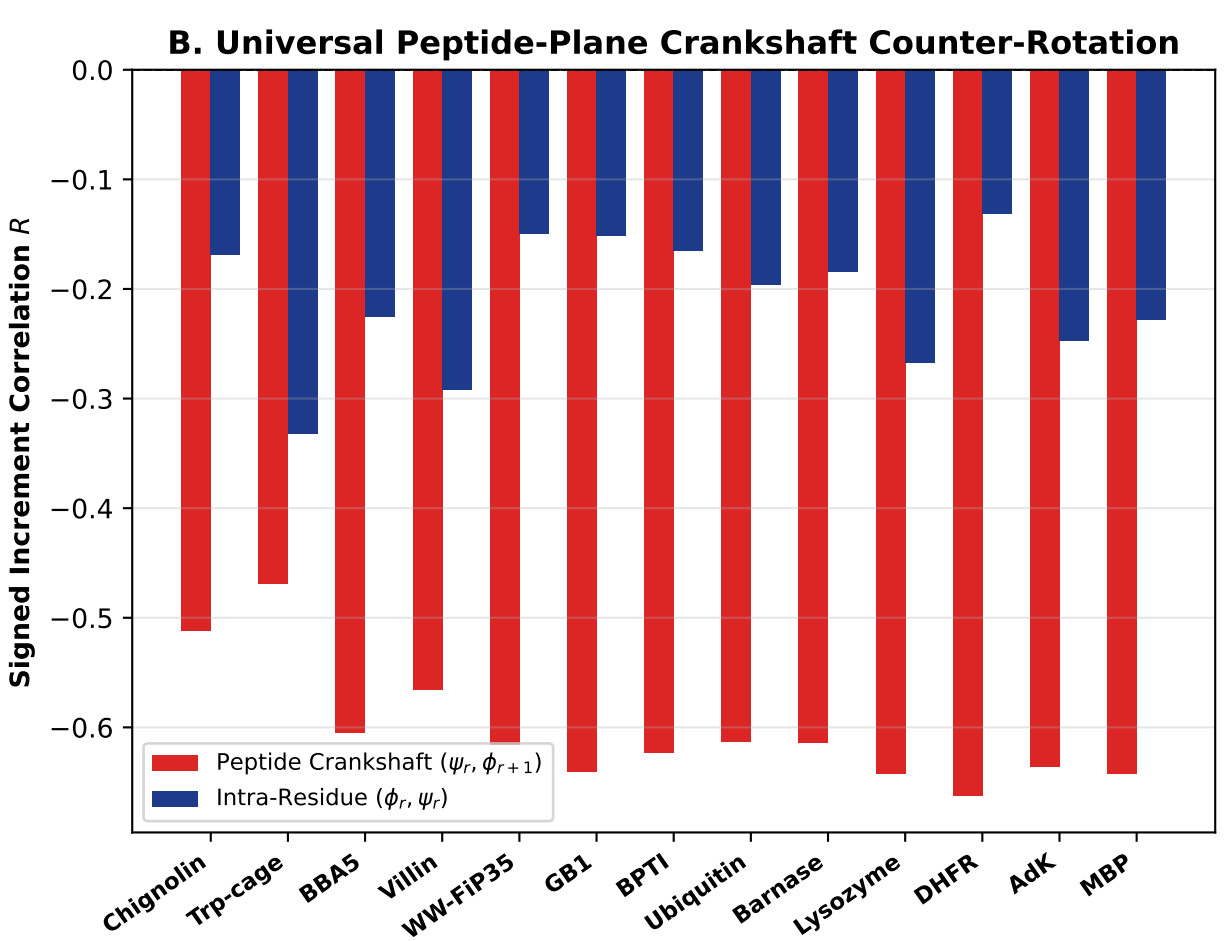
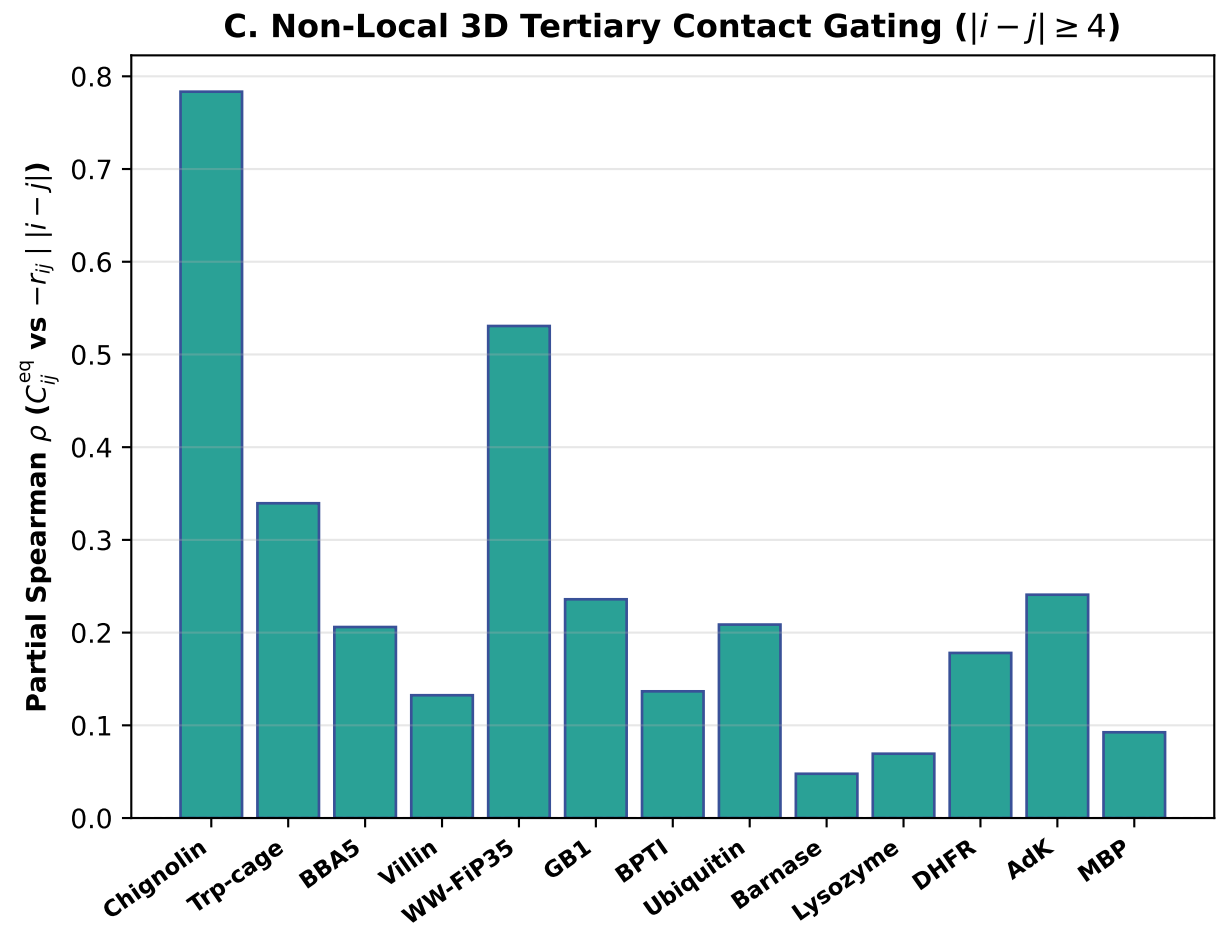
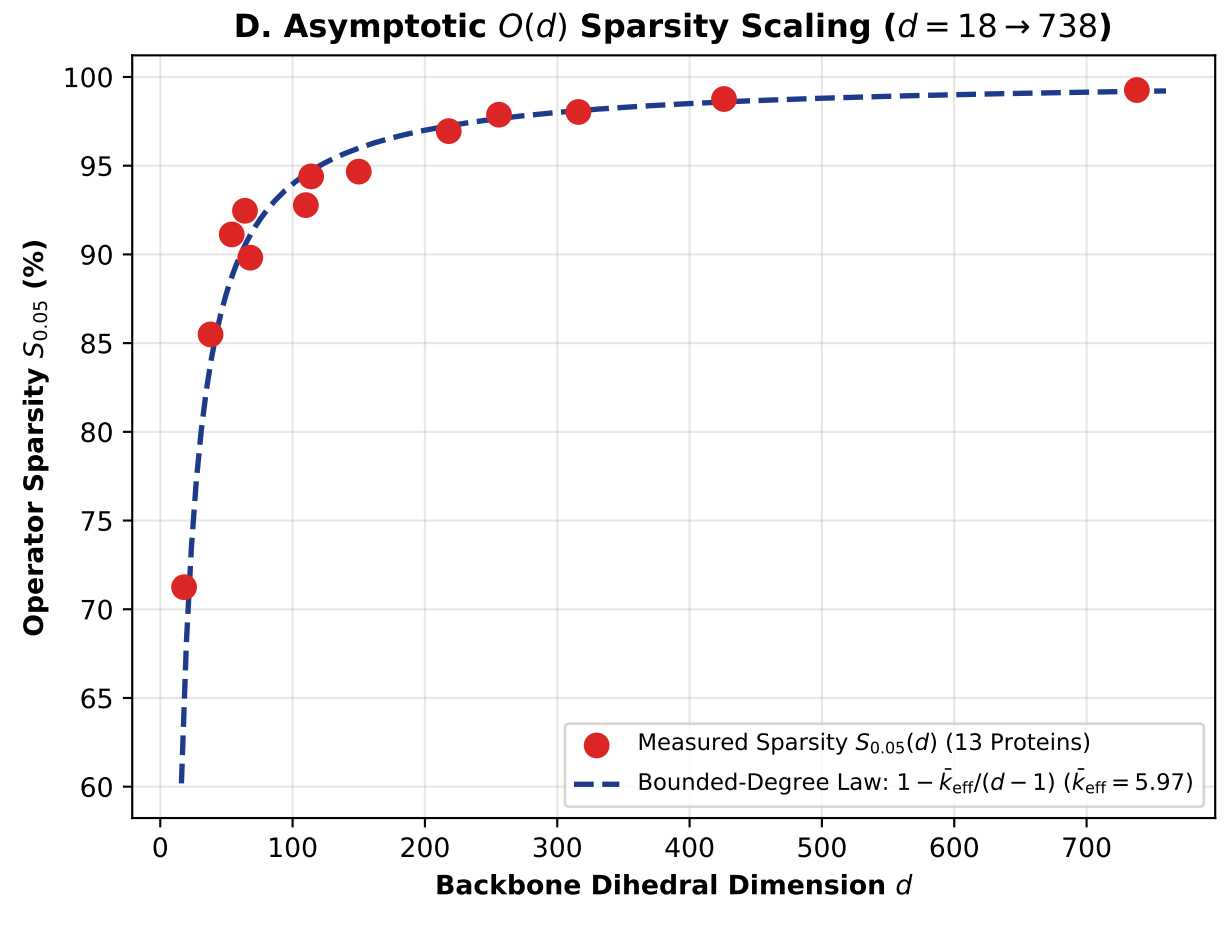

A. Universal Covalent Sequence Decay ($d = 18 \to 738$)
Chignolin ($d = 18$)
Trp-cage ($d = 38$)
Villin ($d = 64$)
Ubiquitin ($d = 150$)
Lysozyme ($d = 256$)
DHFR ($d = 316$)
AdK ($d = 426$)
MBP ($d = 738$)
Mean $|R(\Delta)|$
10^{-1}
10^{-2}
10^{0}
10^{1}
Covalent Dihedral Separation Δ


B. Universal Peptide-Plane Crankshaft Counter-Rotation
Signed Increment Correlation R
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
Peptide Crankshaft (ψ_r, ϕ_{r+1})
Intra-Residue (ϕ_r, ψ_r)
Chignolin
Trp-cage
BBA5
Villin
WW-FiP35
GB1
BPTI
Ubiquitin
Barnase
Lysozyme
DHFR
AdK
MBP


C. Non-Local 3D Tertiary Contact Gating ($|i - j| \geq 4$)
Partial Spearman ρ (C_{ij}^{eq} vs $-r_{ij}$ | $|i - j|$)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Chignolin
Trp-cage
BBA5
Villin
WW-FiP35
GB1
BPTI
Ubiquitin
Barnase
Lysozyme
DHFR
AdK
MBP


D. Asymptotic $O(d)$ Sparsity Scaling ($d = 18 \to 738$)
Operator Sparsity $S_{0.05}$ (%)
100
95
90
85
80
75
70
65
60
Measured Sparsity $S_{0.05}(d)$ (13 Proteins)
Bounded-Degree Law: $1 - \bar{k}_{\text{eff}}/(d-1)$ ($\bar{k}_{\text{eff}} = 5.97$)
0
100
200
300
400
500
600
700
Backbone Dihedral Dimension d